INCH-POUND
MIL-DTL-45499D
3 August 2023
SUPERSEDING
MIL-T-45499C
19 March 1993

DETAIL SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M1A2 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the M1A2 Cartridge Actuated Thruster Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS28741	-	Hose Assembly, Detachable End Fitting, Medium Pressure
MIL-T-45498	-	Thruster, Cartridge Actuated, M1A2 Metal Parts Assembly
MIL-T-50356	-	Thruster, Cartridge Actuated, M1A2 Remanufactured Metal Parts Assembly

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NSWC IHD DRAWINGS

8593284	-	Pin, Shear
8593857	-	Key
8593859	-	Body

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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8593860 - Trunnion

8593862 - Retainer, Cartridge

8593863 - Guide, Firing Pin

8593864 - Pin, Firing

8593867 - Piston

8593869 - Sleeve End

8593870 - Head

8593871 - Spring, Piston Locking

8593885 - Cap, Screw

8593889 - Cartridge, Impulse, M42A1 Assembly

(Copies of these drawings are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Document Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

NATIONAL AEROSPACE STANDARDS COMMITTEE (NA/NAS)

NASM35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series

(Copies of this document are available online at www.aia-aerospace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Government design activity prior to manufacture.

3.2 Components.

3.2.1 Primary components. For the purpose of this specification, the M42A1 impulse cartridge (NSWC IHD Drawing 8593889) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of thrusters; however, one primary component production lot may be used in more than one thruster production lot.

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3.2.2 Component parts. All component parts shall meet the applicable requirements of MIL-T-45498 or MIL-T-50356 prior to assembly.

3.3 Torque. Torque requirements shall be as specified in table I. No movement shall occur upon application of the breakaway torque.

TABLE I. Torque requirements.

Components	A	B	C
	Run on, minimum (inch pounds)	Seating or bottoming (including A) (inch pounds)	Breakaway, minimum (inch pounds)
Head and cartridge retainer	40	300±20	200
Head and body	40	600±20	400

3.4 Trunnion rotation. Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.4.1 Structural strength. The assembled thruster shall withstand a static tensile force of 2,000 pounds minimum, for a minimum of 15 seconds between mounting points without unlocking and without mechanical failure (see 6.5).

3.4.2 End sleeve. Upon completion of assembly and after the structural strength test, the end sleeve shall not rotate.

3.5 Residual magnetism. The thruster shall not deflect the indicator of the compass more than 5 degrees in either direction (see 4.3.2).

3.6 Radiographic examination. There shall be no observable imperfections in components or the assembly when radiographically examined in accordance with 4.2.2).

3.7 Functional.

3.7.1 Load. The thruster, loaded with an acceptable lot of cartridges and fired under load (see 6.3), shall perform as follows: with the piston connected to a 20-pound weight, the piston shall push the weight along a horizontal path of travel, and simultaneously the moving piston and weight shall overcome restraining forces of 1,000 pounds, 6,000 pounds, 3,200 pounds, and 1,000 pounds, with the restraining forces being located from the initial point of travel, a distance of 0 inches, 0.25 inch, 0.40 inch, and 0.75 inch, respectively. After completely overcoming the restraining forces, the piston shall complete its stroke of 2 inches, minimum.

3.7.2 Temperature. The thruster shall comply with all the ballistic requirements throughout the temperature range of −65 to +200 °F.

3.7.3 Initiation. The thruster shall be fired by a source delivering an output pressure of 700 +25, −0 pounds per square inch gage (psig) at the end of 15 feet of aircraft hose, MS28741-4, at 70±5 °F.

3.7.4 Misfires. There shall be no misfires.

3.7.5 Mechanical failures. The thruster shall function without mechanical failure (see 6.5).

3.8 Workmanship. Thrusters shall be free of cracks, splits, or other defects that might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr that might interfere with the assembly or function of the item or that might be injurious to personnel handling the item.

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3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examinations of 4.2 and the tests of 4.3.

4.1.1 Lot. Unless otherwise specified (see 6.2), formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4. The production lot size shall be as specified (see 6.2).

4.2 Examinations.

4.2.1 Visual examination. Visual examination shall be completed on the M1A2 Assembly at a maximum of four power (4×) magnification. Defects such as burrs, cracks, splits, cold shuts, inclusions, and porosity are cause for rejection. The thruster assembly shall be visually inspect to ensure that no foreign material such as chips, dirt, grease, rust, corrosion, or other foreign matter are present. The threaded input and output ports shall be visually inspected to ensure that no foreign matter is present.

4.2.2 Radiographic examination. All assembled thrusters shall be radiographically examined in accordance with ASTM E1742/E1742M, with the exception that a thruster assembly will be used for the penetrameter. As specified (see 6.2), all radiographs shall be provided to the procuring activity. Any observable imperfections in components or assembly shall be cause for rejection. The thrusters shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All thrusters shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The thrusters shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective thrusters found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Guide, firing pin, NSWC IHD Drawing 8593863 (CAGE 19200)
b. Firing pin, NSWC IHD Drawing 8593864 (CAGE 19200)
c. Retainer cartridge, NSWC IHD Drawing 8593862 (CAGE 00000)
d. Cartridge M42A1, NSWC IHD Drawing 8593889 (CAGE 19200)
e. Pin, shear, NSWC IHD Drawing 8593284 (CAGE 19200)
f. Piston, NSWC IHD Drawing 8593867 (CAGE 00000)
g. Cap, screw, NSWC IHD Drawing 8593885 (CAGE 19200)
h. Spring, piston locking, NSWC IHD Drawing 8593871 (CAGE 19200)
i. 4-Keys, NSWC IHD Drawing 8593857 (CAGE 00000)
j. Washer, MS35338-136 (of NASM35338)
k. Trunnion, NSWC IHD Drawing 8593860 (CAGE 00000)
l. Head, NSWC IHD Drawing 8593870 (CAGE 00000)
m. Body, NSWC IHD Drawing 8593859 (CAGE 00000)
n. Sleeve end, NSWC IHD Drawing 8593869 (CAGE 00000)

(As the CAGE code for a drawing never changes, the original CAGE codes for the drawings are listed. The current design activity for all drawings is 14083.)

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4.3 Tests.

4.3.1 Breakaway torque.

4.3.1.1 Head and cartridge retainer. Each head and cartridge retainer subassembly shall be subjected to a breakaway torque. A torque wrench, set at the torque requirements specified in table I, shall be used in conjunction with a fixture to determine the adequacy of the joints. Failure of any subassembly to comply with the requirements of column C in table I shall be cause for rejection of the subassembly.

4.3.1.2 Head and body. Each head and body assembly shall be subjected to a breakaway torque. Failure of any assembly to comply with the requirements of column C in table I shall be cause for rejection of the assembly.

4.3.2 Residual magnetism. Each assembly shall be subjected to a test for excessive residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in the north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly for four full passes. Failure of any unit to comply with 3.5 shall be cause for rejection of the assembly.

4.3.3 Structural strength. Each thruster shall be subjected to a structural strength test. The thruster shall be tested in a tensile testing machine or similar device by the application of the static tensile load between the trunnion lugs and the end sleeve. Failure of any unit to comply with the structural strength requirement of 3.4.1 shall be cause for rejection of the thruster.

4.3.4 Functional tests. The following functional tests shall be performed.

4.3.4.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.7. The test sample shall be rigidly supported so that the entire thrust is borne by the thruster mounting points.

4.3.4.2 Temperature conditioning. The units shall be temperature conditioned as provided in table II. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures. Each assembled thruster shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning thrusters at −65 °F, all condensation shall be removed from the exterior surfaces before returning the thruster to the conditioning chamber.

TABLE II. Conditioning temperature.

Sample size	Method	Temperature (°F)
7	Load (see 3.7.1)	−65 to −70
7	Load (see 3.7.1)	+65 to +75
7	Load (see 3.7.1)	+195 to +205

4.3.4.3 Functional test failure. Failure of any unit to comply with 3.7.1 through 3.7.5 shall be cause for rejection of the lot.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in military Aircrew Emergency Escape Systems. The thruster assembly is a human-rated, aviation critical safety item with no commercial equivalent or substitute.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Formation and identification of lots if other than specified and lot size (see 4.1.1).

c. Directions for shipping radiographs of the entire production lot along with the ballistic sample to the procuring activity (see 4.2.2).

d. Packaging requirements (see 5.1). Recommended packaging requirement is per SPI F00-891-5488.

e. Applicable safety precaution requirements (see 6.6).

6.3 Load firing. The expected performance of the thruster, under conditions simulating those existing in aircraft, is obtained when the thruster satisfactorily meets the conditions of the "load" firing.

6.4 Hazard notice. The thruster described herein and certain of its components are flammable, toxic, or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.5 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6 Safety precautions. When specified (see 6.2), the safety precaution requirements of the "DoD Contractor's Safety Manual for Ammunition, and Explosives," DoD Manual 4145.26 are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Safety Ashore," OP 5 are applicable.

6.7 Subject term (key word) listing.

Ejection systems, aircrew

Emergency escape systems, aircrew

Escape systems, aircrew

Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Navy – OS
Air Force –170

Preparing activity:
Navy – OS
(Project 1377-2023-001)

Review activity:
Air Force – 84

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